



Attitude and Acceleration Autopilot Design

Antiship missile based on the 6DOF course model

Team

Gabriel Henrique Alves Rezende
João Pedro Chamhum Basílio
Mateus Pereira Alves
Matheus Marinheiro Spontam
Sávio Lima Rodrigues

Date: June 25, 2026



Contents

1	Source Code and Adaptation	1
2	Autopilot Description	2
2.1	Attitude Autopilot	2
2.2	Acceleration Autopilot	2
2.3	Actuator and Airframe Data	2
3	Gain Design	3
3.1	Root-Locus Plots	4
4	Step Responses	7
4.1	Attitude Response	7
4.2	Acceleration Response	7
5	Requirement Check	10
6	Engagement Check	11



1 Source Code and Adaptation

The design was based on the folder 6DOF - CURS0/03-autopiloto. The main script used as reference was `ProjRLocus.m`, which designs the attitude and acceleration autopilots by root locus using the short-period transfer functions generated by `FTransDin.m`.

For this work, the original script was copied to `ProjRLocus_antiship.m`. The structure of the professor's code was preserved. The changes were limited to:

- replacing the generic missile data by the antiship missile data from the previous 6DOF work;
- using the DATCOM aerodynamic matrix `M_aed.mat` generated for the antiship missile;
- setting the design condition to Mach 0.9 and altitude 10 m;
- using representative mass/CG points along the burn;
- exporting the figures and the numerical requirement check.

The Simulink file `Autopiloto.slx` was kept in the code folder as the implementation model from the course material. The gain design itself follows the MATLAB root-locus script, which is the part of the professor's code intended for generating the autopilot tables.

```
% Antiship missile design condition used in the assignment.  
altitude = 10;  
Mach = 0.9;  
VetTmp = [0 D.TqB/2 D.TqB D.TqB + D.TqS/2];
```

The selected points cover launch mass, intermediate booster burn, booster burnout and a representative cruise point during the sustainer burn.



2 Autopilot Description

Two autopilots were designed with the same architecture used in the course script.

2.1 Attitude Autopilot

The attitude autopilot uses an inner pitch-rate loop and an outer attitude loop. The inner loop feeds back q , measured by the gyro, and acts through the actuator and the short-period dynamics. The outer loop compares the attitude command with θ , and the plant from q to θ includes the kinematic integrator $1/s$.

$$\theta_c \rightarrow K_\theta \rightarrow (K_q, \text{ actuator}, q/\delta) \rightarrow \frac{1}{s} \rightarrow \theta$$

The inner gain K_q was selected from the root locus of q/δ . The outer gain K_θ was selected from the attitude-loop root locus, using the damping criterion from the course code.

2.2 Acceleration Autopilot

The acceleration autopilot keeps the same inner q -feedback loop. The external loop controls lateral acceleration A_z , with an integrator in the external amplifier, as used in the course material:

$$A_{z,c} \rightarrow \frac{K_a}{s} \rightarrow (K_q, \text{ actuator}, q/\delta) \rightarrow A_z/\delta \rightarrow A_z$$

The first value of K_a was obtained from the acceleration root locus. A final adjustment was then made around that value to keep the acceleration loop stable while satisfying the minimum frequency-response requirement.

2.3 Actuator and Airframe Data

The actuator was kept as the second-order model from `ProjRLocus.m`, with natural frequency 100 Hz and damping $\sqrt{2}/2$. The lateral acceleration limit used in the missile data file is $5g$, and the fin deflection limit used by `DadosControle.m` is 10° .



3 Gain Design

The gain schedule was computed at Mach 0.9, 10 m altitude, and four mass/CG points. The internal gain is the same for the attitude and acceleration autopilots because both use the same q -feedback loop.

Table 1: Computed gain schedule.

t (s)	X_{CG} (m)	K_q attitude	K_θ	K_q accel.	K_a
0.000	2.8325	0.6000	9.3333	0.6000	0.0250
1.725	2.8934	0.6667	10.6667	0.6667	0.0250
3.450	2.9647	0.7333	12.0000	0.7333	0.0250
115.350	2.6012	0.3333	8.0000	0.3333	0.0250

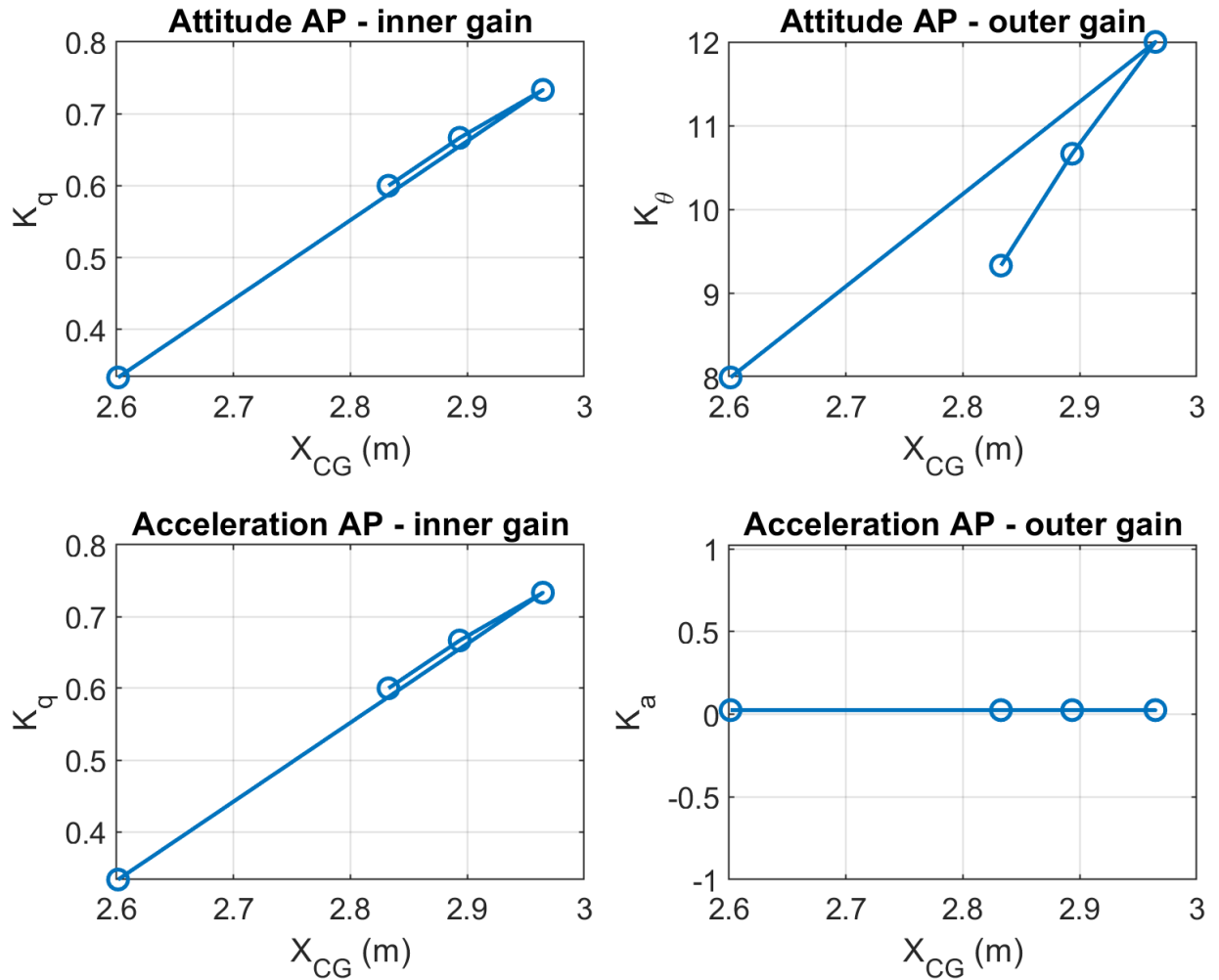


Figure 1: Internal and external gains as a function of CG position.

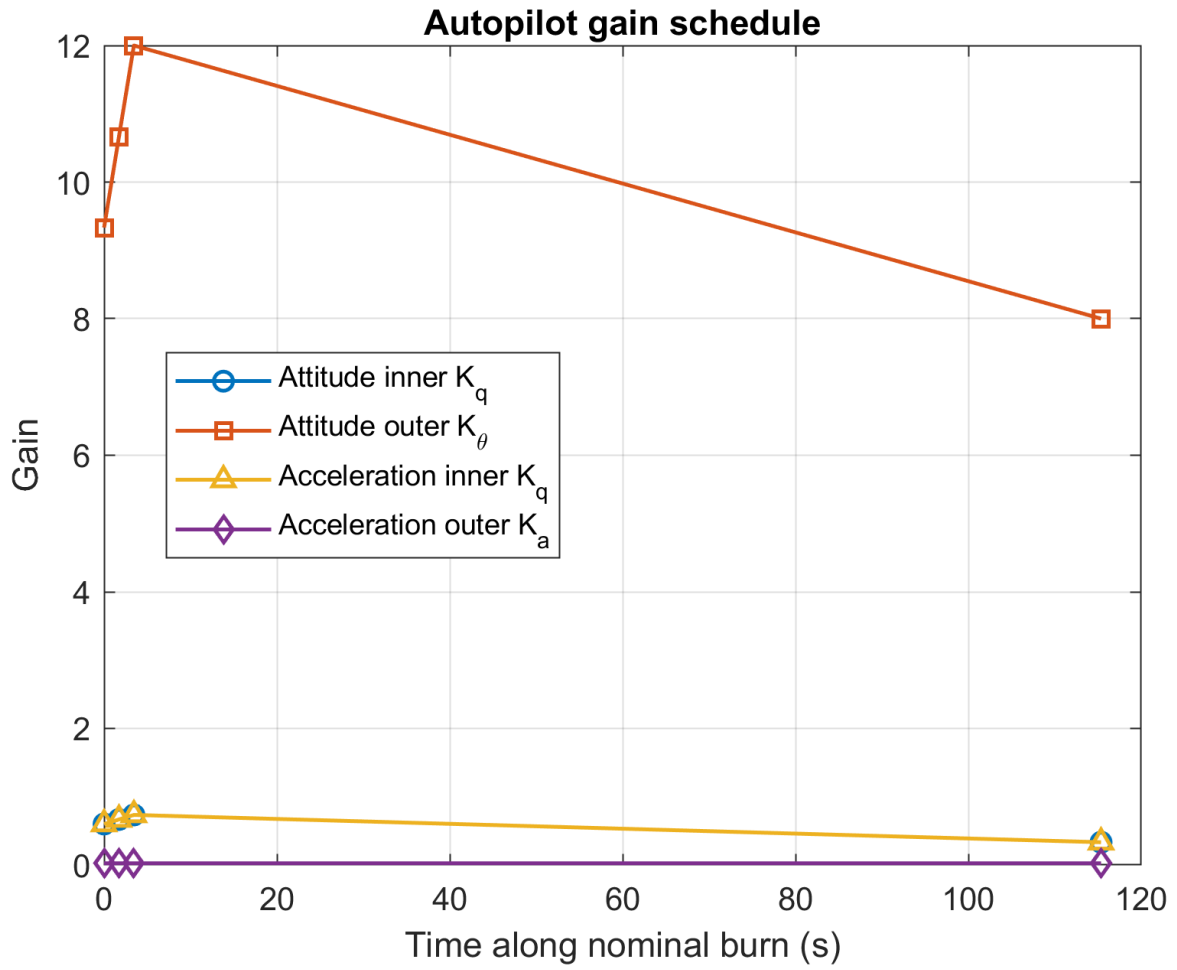


Figure 2: Gain schedule along the representative burn points.

3.1 Root-Locus Plots

The following root-locus plots show the nominal design point, taken during the sustainer phase. The internal loop is common to both autopilots.

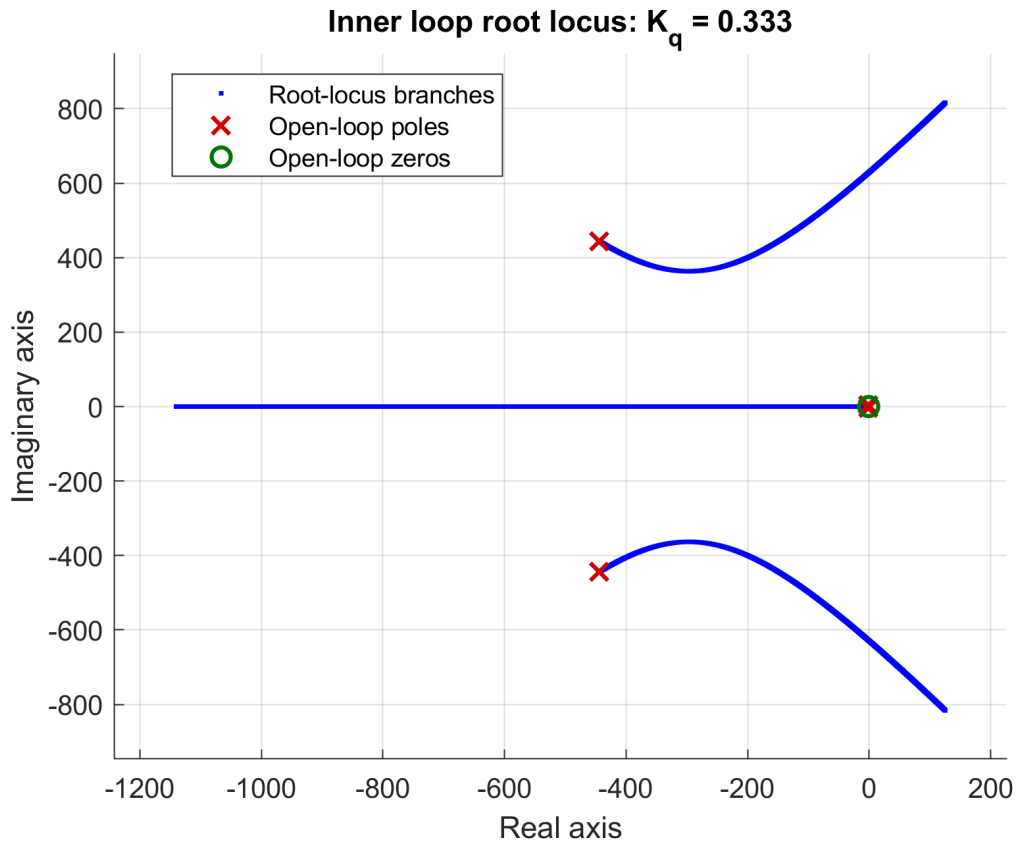


Figure 3: Root locus of the internal attitude loop.

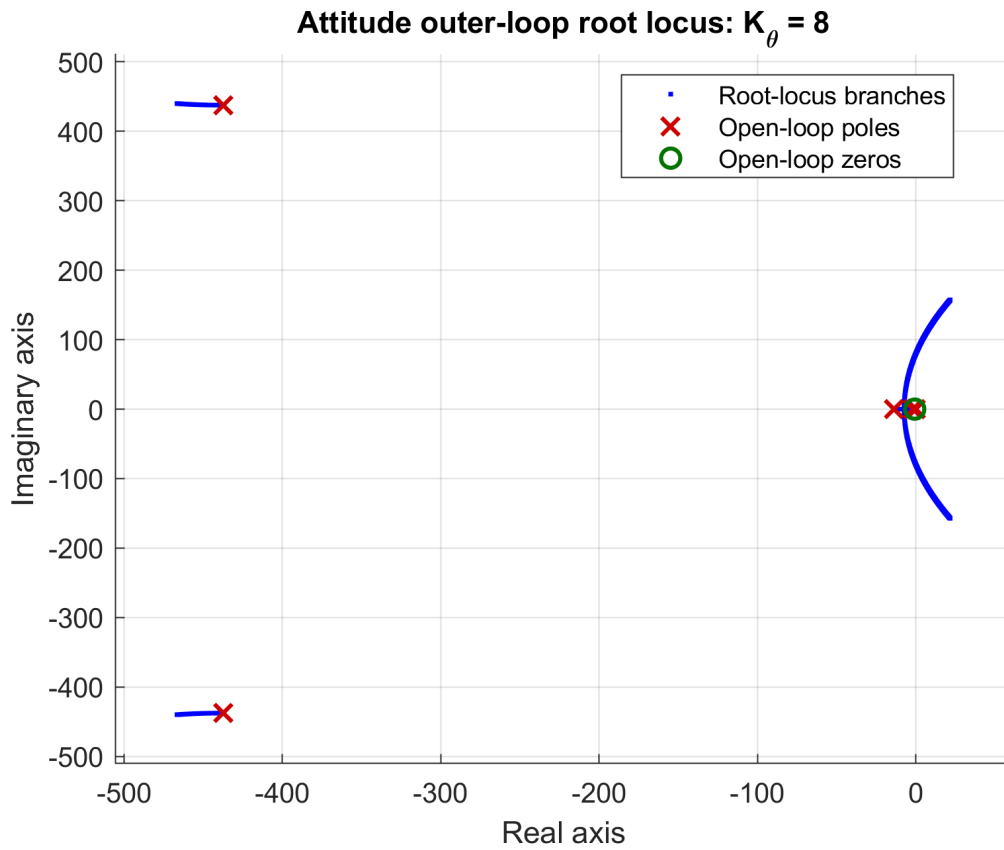


Figure 4: Root locus of the external attitude loop.

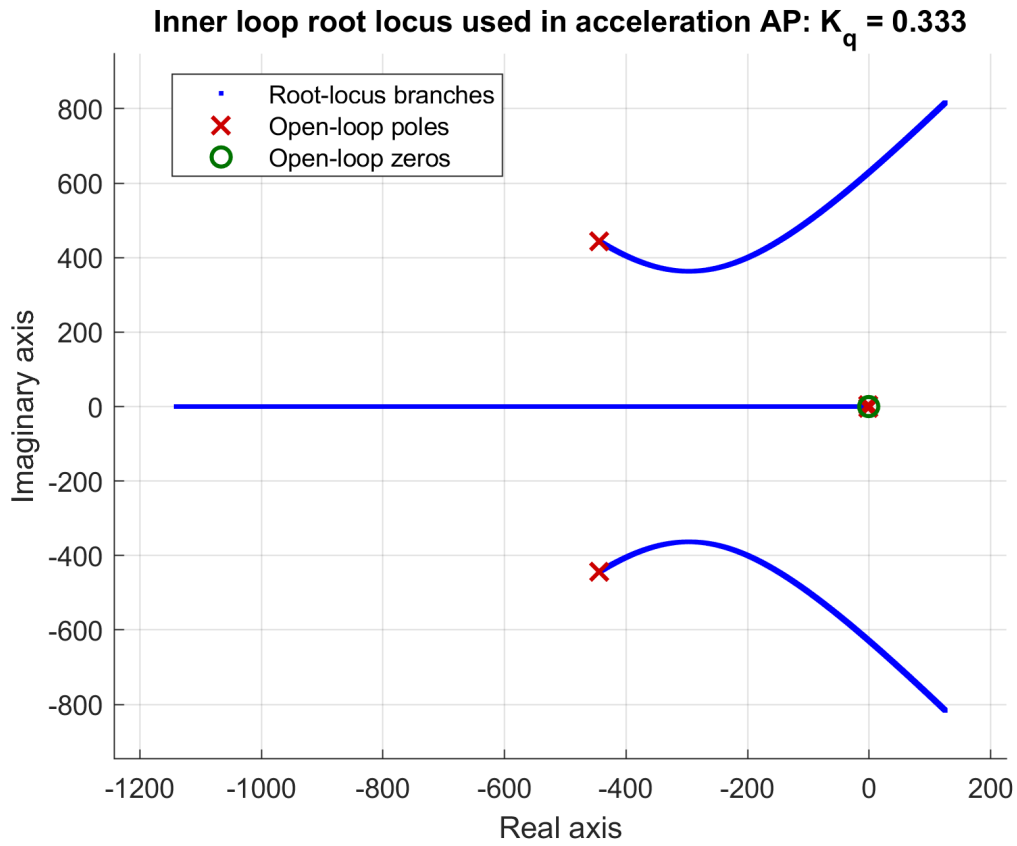


Figure 5: Root locus of the internal loop used by the acceleration autopilot.

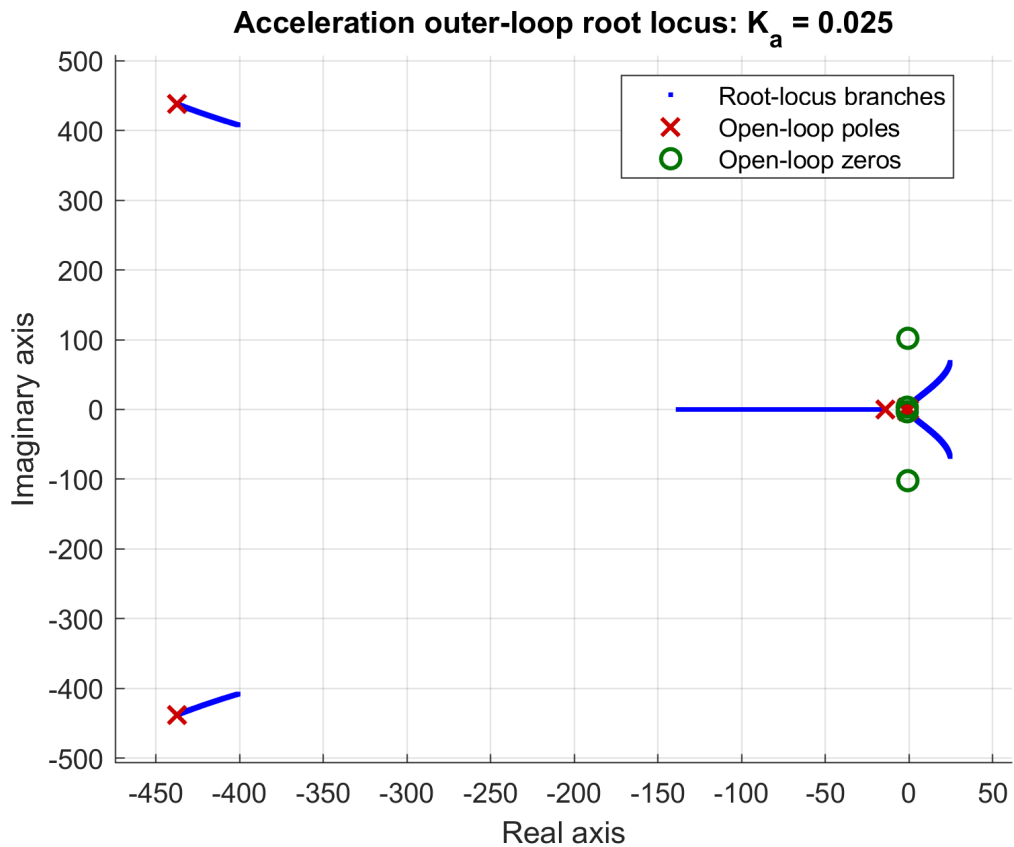


Figure 6: Root locus of the external acceleration loop.



4 Step Responses

The closed-loop step responses were generated at the nominal sustainer point:

$$X_{CG} = 2.6012 \text{ m}, \quad m = 556.921 \text{ kg}, \quad I_y = 1611.097 \text{ kg.m}^2.$$

4.1 Attitude Response

The attitude loop tracks a unit attitude command with small overshoot and unit steady-state gain. The response is initially fast, and the remaining slow component is caused by the lower-frequency closed-loop pole.

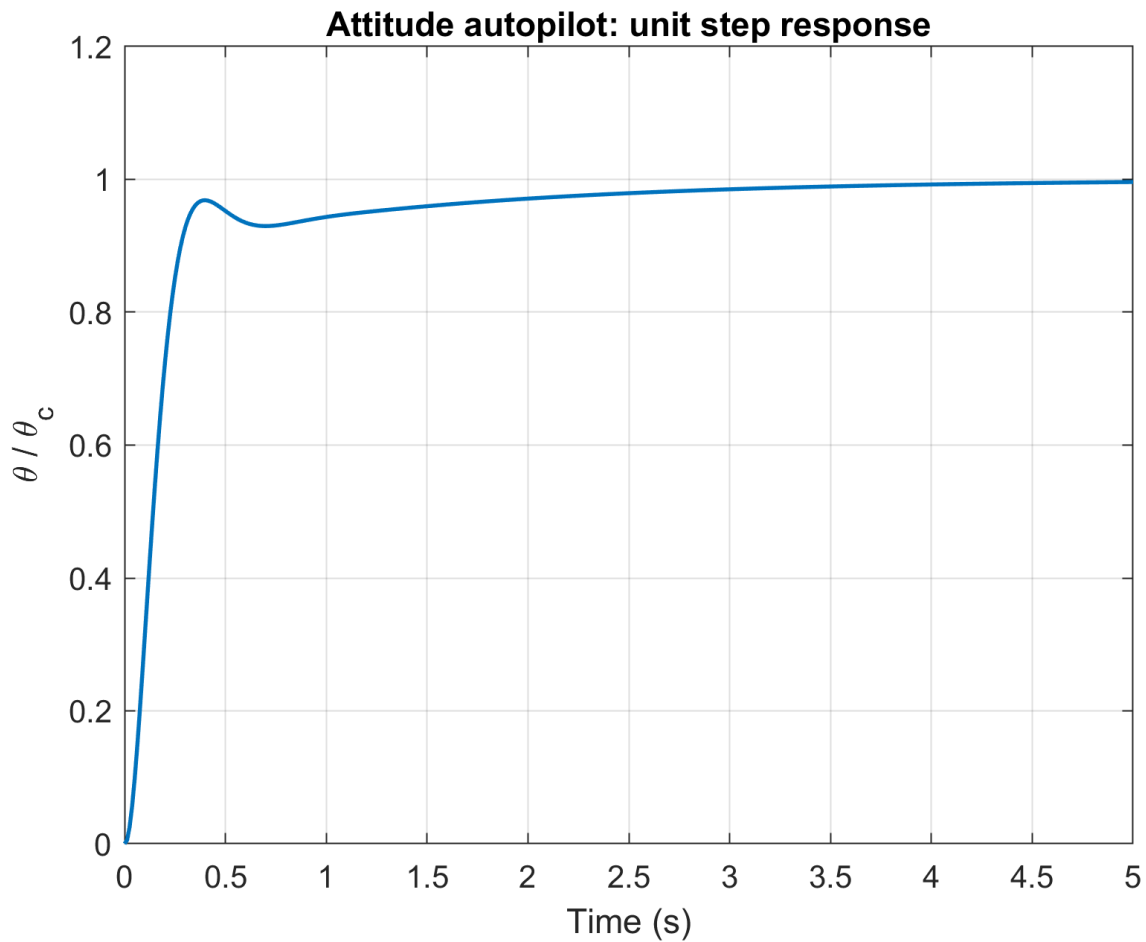


Figure 7: Closed-loop attitude response to a unit step.

4.2 Acceleration Response

The acceleration loop was tested both with a unit command and with a $5g$ command. The response has noticeable overshoot, but it remains stable and converges to the commanded acceleration.

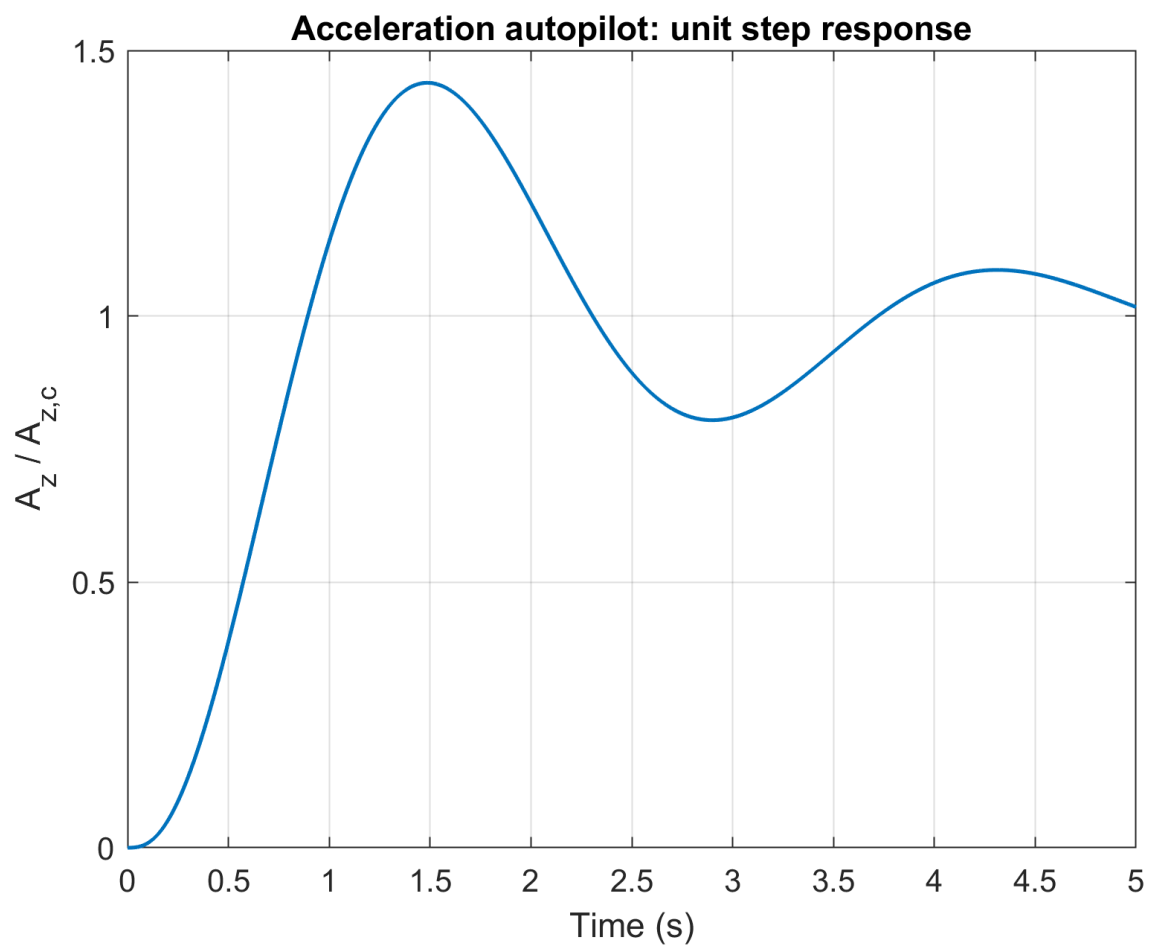


Figure 8: Closed-loop acceleration response to a unit step.

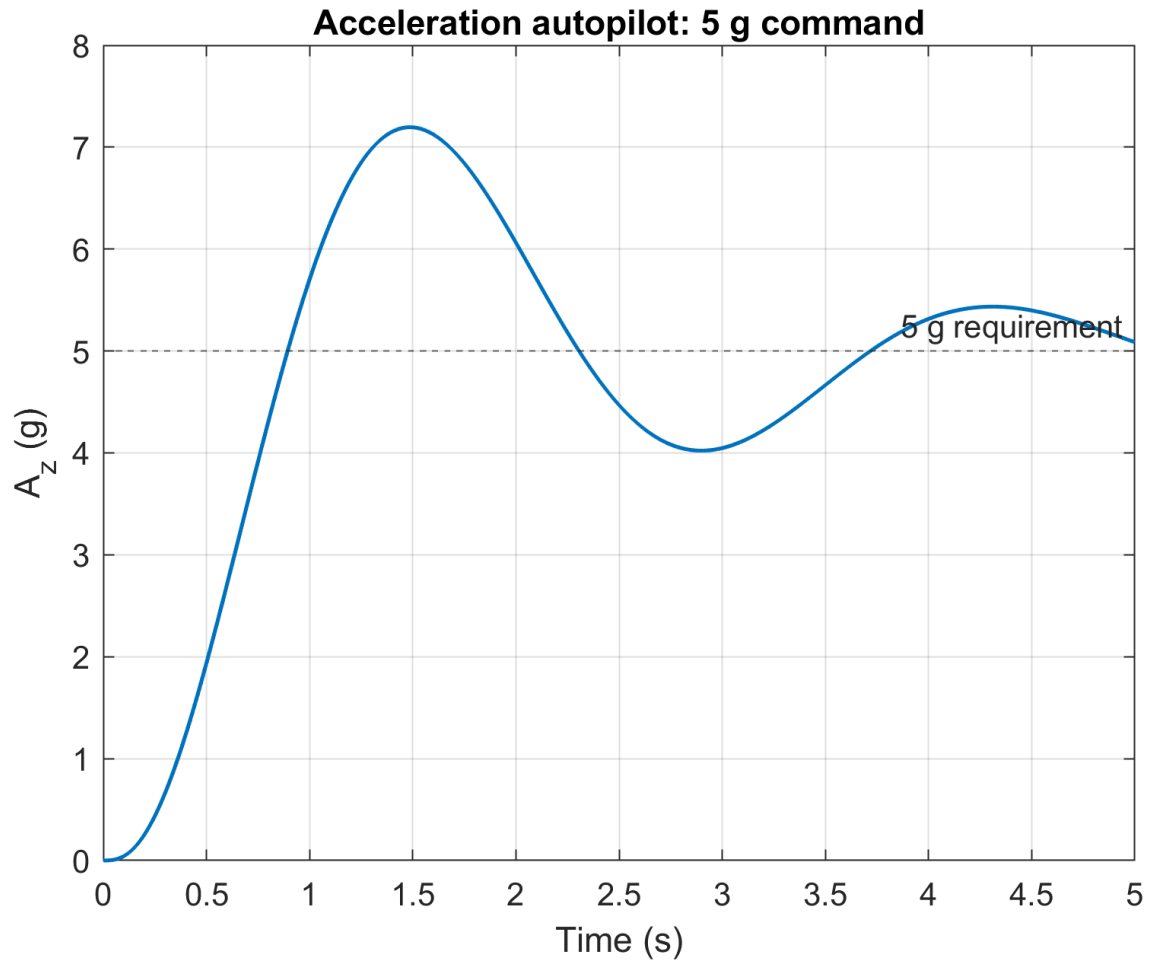


Figure 9: Closed-loop acceleration response to a 5g command.

For the nominal point, the MATLAB step metrics were:

Table 2: Nominal closed-loop step metrics.

Autopilot	Rise time (s)	Settling time (s)	Overshoot (%)	DC gain
Attitude	0.228	2.562	0.0	1.000
Acceleration	0.564	6.218	43.7	1.000



5 Requirement Check

The requirement was checked at Mach 0.9, altitude 10 m, using the nominal sustainer point. The frequency-response requirement was evaluated using closed-loop bandwidth, because it directly measures the usable response frequency of each autopilot.

Table 3: Requirement verification.

Requirement	Result	Limit	Status
Lateral acceleration capability	13.52g	$\geq 5g$	Satisfied
Attitude autopilot bandwidth	1.652 Hz	≥ 0.5 Hz	Satisfied
Acceleration autopilot bandwidth	0.537 Hz	≥ 0.5 Hz	Satisfied

The lateral acceleration capability was estimated from the acceleration transfer function and the 10° fin-deflection limit. The result is above $5g$, so the missile has enough control authority at the design condition.

The acceleration autopilot is the limiting case. The original root-locus point gave an external gain that was too aggressive for the nominal model, so K_a was reduced to 0.025. This keeps the loop stable while preserving a bandwidth slightly above 0.5 Hz. The response is therefore acceptable for the stated requirements, although it is more oscillatory than the attitude loop.



6 Engagement Check

The autopilot design was also tested in a horizontal point-mass engagement. This test uses the same mission geometry adopted in the guidance work and applies the main R4 constraints: $5g$ lateral acceleration limit, seeker cone of 40° , Mach 0.9 at 10 m, and an airframe response with $w_n = 0.5$ Hz. The target is fixed at the origin.

The terminal law is proportional navigation. Before terminal guidance, Case 1 flies straight to the target line and switches at 20 km. Case 2 starts with the seeker on, but the missile first performs a commanded turn until the target is inside the seeker cone.

Table 4: Horizontal engagement results with R4 constraints.

Case	Minimum range	Time at minimum range	Status
1: 65 km, bearing 0°	48.93 m	212.10 s	Satisfied
2: 15 km, bearing 90°	47.06 m	54.43 s	Satisfied

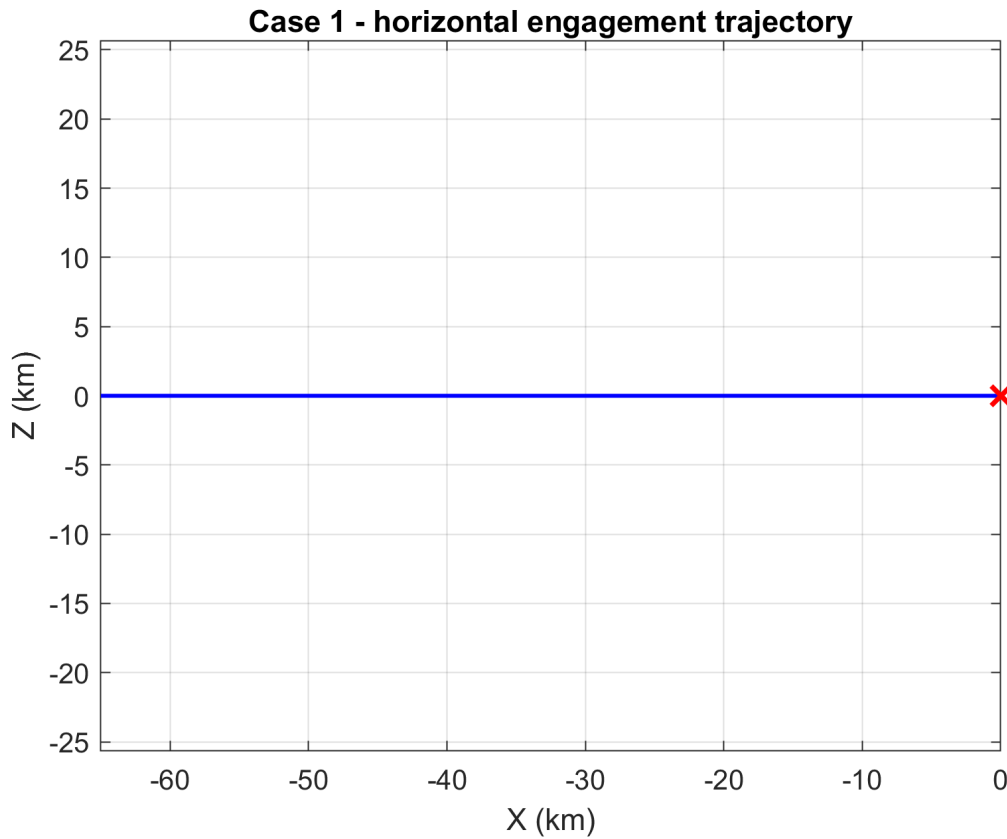


Figure 10: Case 1 horizontal trajectory.

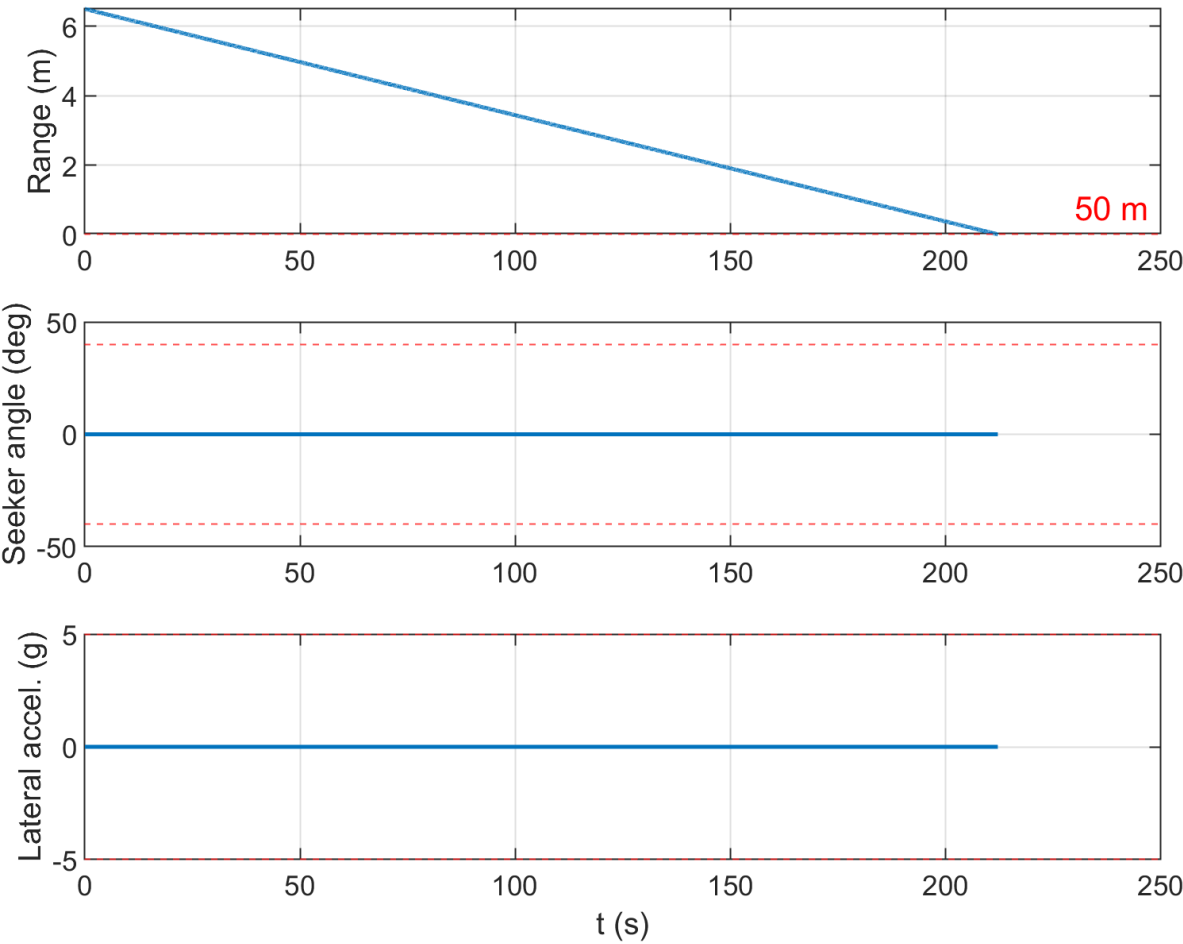


Figure 11: Case 1 range, seeker angle, and lateral acceleration.

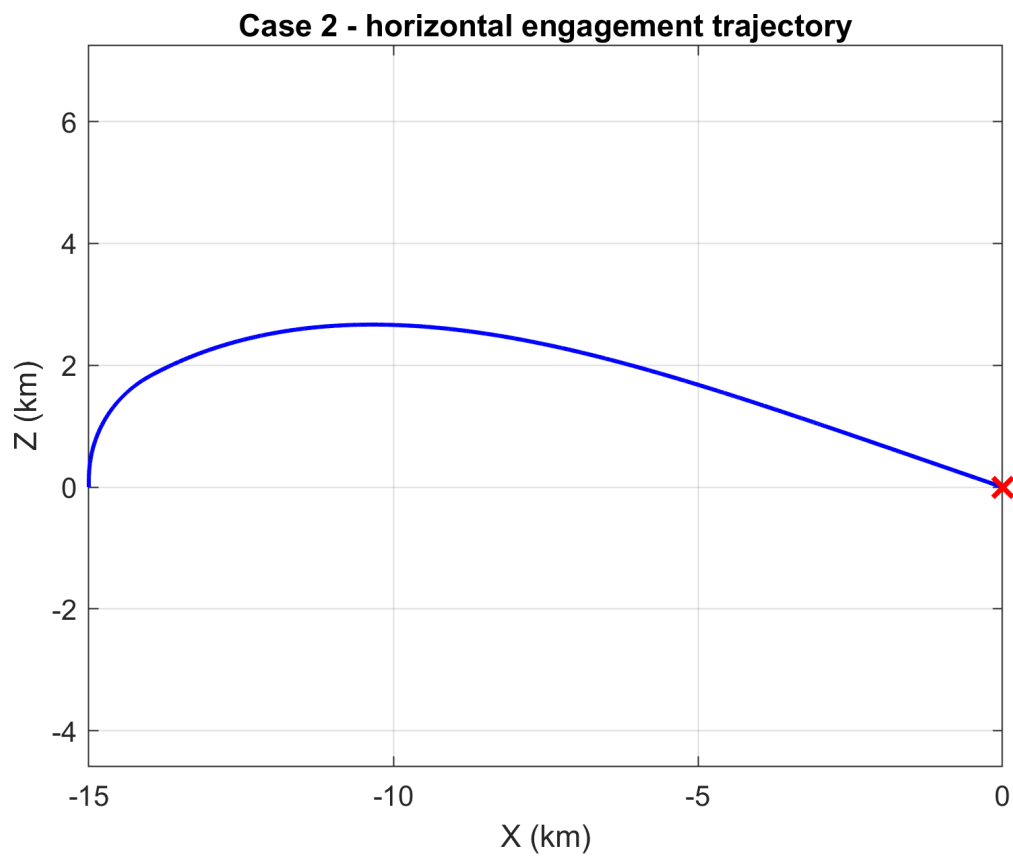


Figure 12: Case 2 horizontal trajectory.

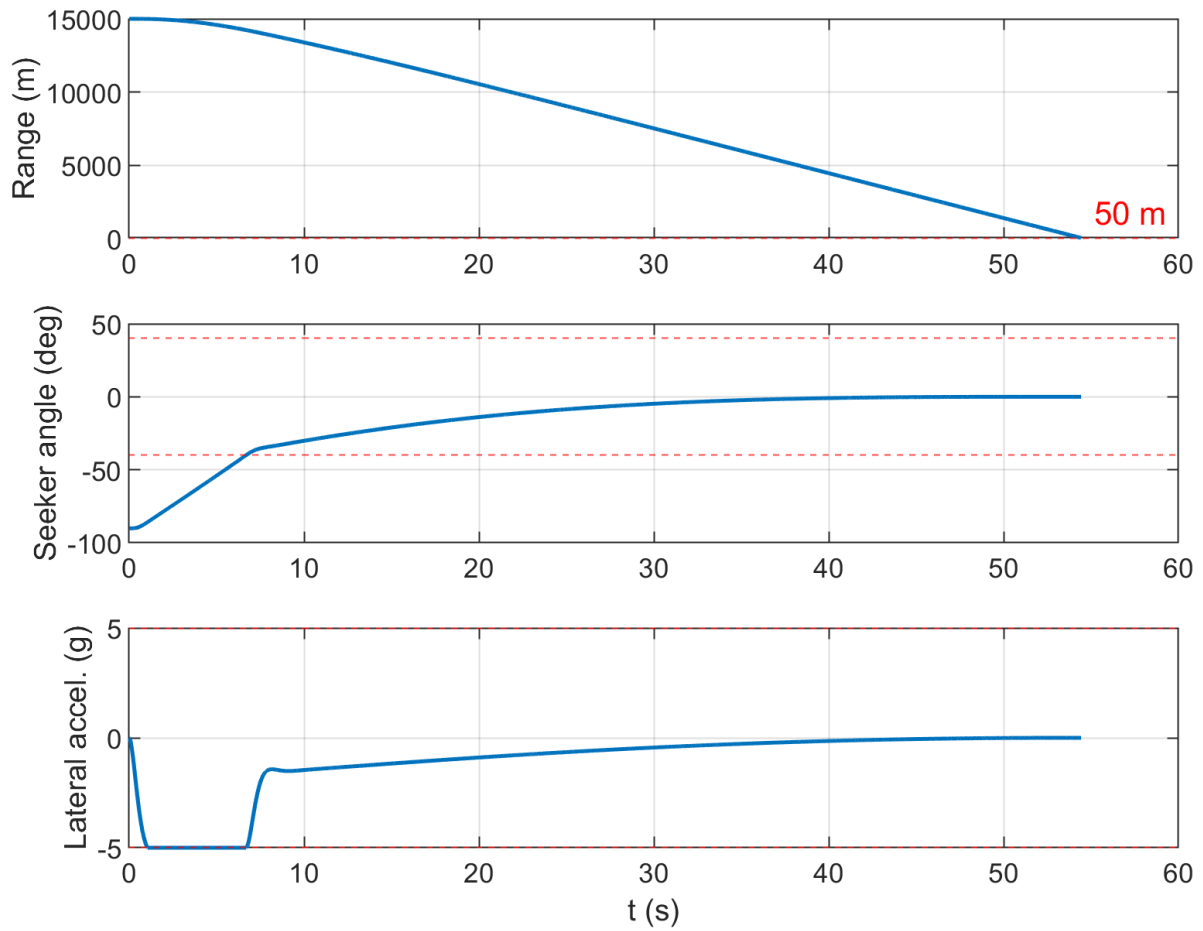


Figure 13: Case 2 range, seeker angle, and lateral acceleration.

Both cases satisfy the 50 m miss-distance criterion. Case 1 is nearly collinear, so it requires almost no lateral acceleration. Case 2 uses the full $5g$ limit during the initial turn, which is consistent with the harder 90° launch geometry.